

The Complex Free Space over $\{0\} \cup \mathbb{T}$ is One-Dimensional

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Source Question

The source is Valentin Ferenczi, Pedro L. Kaufmann, and Eva Pernecká, *Equivariant liftings in Lipschitz-free spaces*, arXiv:2501.06984. In Section 3.2 the authors define, for a pointed metric space M with an isometric action of the unit circle $\mathbb{T}$, the complex quotient

$$Y_{\mathbb{T}} = \overline{\text{span}}\{\delta(e^{i\theta}.m) - \cos \theta \delta(m) - \sin \theta \delta(i.m) : m \in M, \theta \in \mathbb{R}\}$$

and

$$\mathcal{F}_{\mathbb{C}}(M) = \mathcal{F}(M)/Y_{\mathbb{T}},$$

with complex scalar multiplication determined by

$$i(\delta(m) + Y_{\mathbb{T}}) = \delta(i.m) + Y_{\mathbb{T}}.$$

Writing $\delta_{\mathbb{C}} = \pi_{\mathbb{C}}\delta$, the paper then asks:

Question 23. Assume $\{0\} \cup \mathbb{T}$ is equipped with the natural action of $\mathbb{T}$. Is the space $\mathcal{F}_{\mathbb{C}}(\{0\} \cup \mathbb{T})$ isomorphic to $L_1([0, 1], \mathbb{C})$?

For Banach spaces, a different approach would be needed to investigate the question above.

QUESTION 23. Assume $\{0\} \cup \mathbb{T}$ is equipped with the natural action of $\mathbb{T}$. Is the space $\mathcal{F}_{\mathbb{C}}(\{0\} \cup \mathbb{T})$ isomorphic to $L_1([0, 1], \mathbb{C})$?

The space $\mathcal{F}(\{0\} \cup \mathbb{T})$ is linearly isomorphic to $L_1 = L_1([0, 1], \mathbb{R})$ (to see this, combine [2, Lemma 2.8 (iii) and Theorem 4.21] with [13]). So what we know about the above question is

Answer

Under the natural realization of $\{0\} \cup \mathbb{T}$ as a pointed subset of the complex plane and the action $e^{i\theta}.z = e^{i\theta}z$, the answer is negative. More precisely, $\mathcal{F}_{\mathbb{C}}(\{0\} \cup \mathbb{T})$ is complex one-dimensional.

Proof Intuition

The quotient defining $\mathcal{F}_{\mathbb{C}}(M)$ is designed to force the circle action to agree with complex scalar multiplication. When $M = \{0\} \cup \mathbb{T}$, the whole nonzero part of M is a single $\mathbb{T}$ -orbit. Therefore every generator $\delta_{\mathbb{C}}(e^{i\theta})$ is forced to equal the scalar multiple $e^{i\theta}\delta_{\mathbb{C}}(1)$. Thus the complex free space has only one possible generator. The source paper's Observation 21 then says this generator is not zero when $M \subset \mathbb{C}$ carries the inherited action.

Theorem and Proof

Theorem 1. *Let $M = \{0\} \cup \mathbb{T} \subset \mathbb{C}$, pointed at 0, with its usual metric and the usual circle action $e^{i\theta}.z = e^{i\theta}z$. Then $\mathcal{F}_{\mathbb{C}}(M)$ is complex-linearly isomorphic to $\mathbb{C}$. In particular, $\mathcal{F}_{\mathbb{C}}(\{0\} \cup \mathbb{T})$ is not isomorphic to $L_1([0, 1], \mathbb{C})$.*

Proof. Let

$$u = \delta_{\mathbb{C}}(1) = \delta(1) + Y_{\mathbb{T}} \in \mathcal{F}_{\mathbb{C}}(M).$$

The defining relation for $Y_{\mathbb{T}}$, applied to $m = 1$, gives for every $\theta \in \mathbb{R}$

$$\delta_{\mathbb{C}}(e^{i\theta}) = \cos \theta \delta_{\mathbb{C}}(1) + \sin \theta \delta_{\mathbb{C}}(i).$$

By the definition of the complex structure on $\mathcal{F}_{\mathbb{C}}(M)$, we have

$$\delta_{\mathbb{C}}(i) = i \delta_{\mathbb{C}}(1) = iu.$$

Therefore

$$\delta_{\mathbb{C}}(e^{i\theta}) = \cos \theta u + \sin \theta iu = e^{i\theta}u.$$

Also $\delta_{\mathbb{C}}(0) = 0$, since 0 is the base point of the Lipschitz-free space. Hence all generators $\delta_{\mathbb{C}}(M)$ lie in the one-dimensional complex subspace $\mathbb{C}u$. Because $\mathcal{F}_{\mathbb{C}}(M)$ is the closed complex linear span of $\delta_{\mathbb{C}}(M)$, it follows that

$$\dim_{\mathbb{C}} \mathcal{F}_{\mathbb{C}}(M) \leq 1.$$

It remains only to rule out the zero space. The source paper proves in Observation 21 that if M is a subset of a complex Banach space X and the $\mathbb{T}$ -action on M is induced by the complex scalar multiplication of X , then $\delta_{\mathbb{C}} : M \rightarrow \mathcal{F}_{\mathbb{C}}(M)$ is isometric. Here $M \subset \mathbb{C}$, so this applies. In particular,

$$\|u\| = \|\delta_{\mathbb{C}}(1) - \delta_{\mathbb{C}}(0)\| = |1 - 0| = 1,$$

and $u \neq 0$. Thus $\mathcal{F}_{\mathbb{C}}(M) = \mathbb{C}u$, a one-dimensional complex Banach space.

The Banach space $L_1([0, 1], \mathbb{C})$ is infinite-dimensional, so it cannot be linearly isomorphic to $\mathcal{F}_{\mathbb{C}}(M)$. This gives a negative answer to Question 23 under the natural metric/action interpretation. $\square$

Verification Notes

The dimension collapse is purely algebraic inside the quotient: for any metric on an abstract pointed $\mathbb{T}$ -orbit for which the construction is defined, the same computation gives $\dim_{\mathbb{C}} \mathcal{F}_{\mathbb{C}}(\{0\} \cup \mathbb{T}) \leq 1$. The nonzero conclusion above uses the source paper's Observation 21, hence the natural interpretation that $\{0\} \cup \mathbb{T}$ is the subset of the complex Banach space $\mathbb{C}$ with the induced metric and scalar action.

Novelty and Duplicate Checks

Before promotion, I searched the cheap run indexes for arXiv 2501.06984, the title terms, and core phrases including `equivariant liftings`, `Lipschitz-free spaces`, `complex Lipschitz-free`, `F_C`, `{0} union T`, and `L1([0,1], C)`. No exact packet or attempt hit was present; the hits were adjacent Lipschitz-free-space work.

I also ran bounded web searches for exact and near-exact phrases such as

- " $F_C(\{0\})$ " " $L_1([0,1])$ ",
- "Equivariant liftings in Lipschitz-free spaces" " F_C ",
- "complex Lipschitz-free space" "unit circle",
- "Ferenczi" "Kaufmann" "Pernecká" " F_C ".

The only directly relevant result surfaced was the source arXiv paper itself. The argument is therefore promoted as a claimed, elementary counterexample requiring human review, especially on the intended metric convention.

References

References

- [1] V. Ferenczi, P. L. Kaufmann, and E. Pernecká, *Equivariant liftings in Lipschitz-free spaces*, arXiv:2501.06984, 2025.

Human Review Recommendation

This packet should be reviewed as an erratum-style consequence of the definitions. The key point for review is not the algebraic computation, but whether Question 23 intended a metric model other than the natural subset $\{0\} \cup \mathbb{T} \subset \mathbb{C}$. Under the natural subset-of- $\mathbb{C}$ reading, the source paper's own Observation 21 makes the collapse nondegenerate and the answer is decisively negative.